

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 2.2 REFLECTION AND CONGRUENCE: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Driving question: Why does a matatu's rear-view mirror show an image that is the same size and shape as the real object behind it, yet appears flipped, and how can we use this to prove that the original and its mirror image are always congruent? Remember the matatu conductor on the Nairobi–Thika highway holding a small rectangular mirror so the driver can see a lorry approaching from behind. Use everything you learned in Lessons 1 to 8 to answer each question below in the space under it. Write in full sentences, show all your working for calculations, and draw neat labelled diagrams with a ruler where asked. Use correct words such as mirror line, image, object, perpendicular bisector, congruent, SSS and lateral inversion.

Part 1: What the mirror does to the lorry

On the Nairobi–Thika highway the conductor's mirror shows three things about the lorry's image: (a) it appears as far behind the mirror surface as the real lorry is in front of it, (b) it is the same size as the lorry, and (c) the lorry's right headlight appears on the left. Draw a labelled diagram showing the lorry (object), the mirror line and the image. Then use the properties of reflection you discovered in Lesson 4 to explain each of the three observations (a), (b) and (c).

Diagram: the mirror line is drawn as a straight line. The object (lorry) is on one side and the image is on the opposite side. Dotted lines join corresponding points such as the right headlight and its image. Each dotted line meets the mirror line at 90 degrees with equal marked distances on each side. (a) Property II: the object and image are equidistant from the mirror line, so the image looks as far behind the glass as the lorry is in front of it. (b) Property I: the image is the same size as the object, because reflection does not stretch or shrink any length. We confirmed this when the tracing-paper fold along the mirror line made the object and image coincide exactly. (c) Property IV: the image is laterally inverted, so the lorry's right headlight appears on the left in the mirror. This is a flip of left and right, not a change of size or shape. The mirror line is also the perpendicular bisector of every segment joining an object point to its image point (Property III), so the position of the image is completely predictable.

Part 2: Coordinates and the mirror line

(a) Imagine the matatu mirror is the y-axis of a Cartesian plane. Write the rule that maps any object point (a, b) to its image,

(a) Reflection in the y-axis maps (a, b) to (-a, b). The y-coordinate (the distance up) stays the same and the x-coordinate changes sign, so the point is the same distance from the mirror line but on the opposite side. No distance between points is changed. (b) Step 1, midpoint of PP': $M = ((2+6)/2, (1+5)/2) = (4, 3)$. The midpoint lies on the mirror line because the mirror line bisects PP'. Step 2, gradient of PP' = $(5-1)/(6-2) = 4/4 = 1$. The mirror line is perpendicular to PP', so its gradient is the negative reciprocal, which is -

and explain in one sentence what stays the same and what changes. (b) A point P(2, 1) has the image P'(6, 5) after reflection in an unknown mirror line. Find the equation of the mirror line. Show each step of your working and say which facts about the mirror line you used.	1. Step 3, use $y - y_1 = m(x - x_1)$: $y - 3 = -1(x - 4)$, so $y - 3 = -x + 4$, giving $y = -x + 7$. The mirror line is $y = -x + 7$. I used the facts that the mirror line is the perpendicular bisector of the segment joining an object point to its image.
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Part 3: Proving the image is congruent

Triangle NMK is reflected in a mirror line to give triangle N'M'K'. (a) Explain why the sides NM, MK and KN are equal in length to N'M', M'K' and K'N'. (b) Name the congruence test that proves triangle NMK is congruent to triangle N'M'K', and write the congruence statement. (c) Why do we not need to measure all six elements (three sides and three angles) of the two triangles?	(a) Reflection preserves all distances. Each vertex and its image are the same perpendicular distance from the mirror line, so no length is stretched or shrunk. Therefore $NM = N'M'$, $MK = M'K'$ and $KN = K'N'$. (b) All three pairs of corresponding sides are equal, so the triangles are congruent by SSS. Triangle NMK is congruent to triangle N'M'K'. (c) The four congruence tests (SSS, SAS, AAS/ASA and RHS) give the minimum information needed to guarantee that two triangles are exactly the same shape and size. Once three sides are known to match, the other measurements must match too, so the remaining three angles do not need to be measured.
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Part 4: Why the image looks flipped

A classmate says: 'If the lorry's image is congruent to the lorry, it should be possible to place one exactly on top of the other without lifting it off the page, so the right headlight should not appear on the left.' Respond to your classmate. In your answer, name the type of congruence that a reflection produces and explain how it resolves the puzzle in the driving question.	Your classmate is mixing up size and shape with orientation. The image is congruent to the lorry because all lengths and angles are the same (SSS). However, reflection reverses orientation, so the congruence is opposite (indirect) congruence. The image is a perfect copy that is a mirror copy. To lay it exactly on the object you would have to lift it and flip it over, which is why the lorry's right headlight appears on the left. This lateral inversion is a change of orientation only. It is not a change in size or shape, so the image and object are always congruent even though the image looks flipped. This is the answer to the driving question: the mirror preserves every length and angle (so the image is the same size and shape), and it reverses orientation (so it appears flipped).
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Part 5: Using reflection and congruence in real life

(a) Give two examples from Kenyan daily life where reflection is used or can be seen, and one example where congruence is used or can be seen. (b) Look back at the model you drew in Lesson 1. Describe two ideas in it that you would keep and two things you would change or add now, using at least two terms from this unit.	(a) Reflection: the matatu conductor's mirror helps the driver judge the true size and distance of a lorry behind, which is a road-safety use. Symmetry in Maasai beadwork collars and in the Kenya flag outline are lines of symmetry, which are mirror lines. Congruence: identical triangular samosa pastries cut from the same template, or identical triangular roof trusses on buildings in the Rift Valley, or identical tiles on Nairobi pavements. (b) Sample answer. Keep: the idea that the image looks the same size as the object, and that the mirror is in the middle between object and image. Change or add: the mirror line is the perpendicular bisector of every object-image segment, so I would add right-angle marks and equal-distance arrows; and I would add the coordinate rule (a, b) to (-a, b) for the y-axis and say the image is congruent by SSS with opposite congruence. Any two accurate keeps and two accurate changes using unit vocabulary are acceptable.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Explaining the phenomenon with properties of reflection	Correctly explains equal distance, equal size and lateral inversion, linking each to the matatu mirror and the lorry, with a fully labelled diagram showing perpendicular connectors.	Explains at least two of the three observations correctly using the properties of reflection; the diagram is mostly labelled.	Describes what is seen but does not link it to properties of reflection, or the diagram is missing or unclear.
Coordinate rules and mirror-line equation	States (a, b) to (-a, b) correctly and finds $y = -x + 7$ with all steps shown (midpoint, gradient, negative reciprocal, equation) and a reason for each.	Finds the correct equation with minor slips in working or missing reasons, or states the rule correctly with a small error in the equation.	Cannot state the rule or find the midpoint and gradient correctly, and does not reach the equation.
Congruence reasoning (SSS and opposite congruence)	Explains that reflection preserves distances, names SSS with a correct congruence statement, and explains why six elements are not needed and why the congruence is opposite.	Names SSS and gives a correct statement but the explanation of why distances are preserved or why the congruence is opposite is incomplete.	Does not name a valid congruence test or confuses congruence with orientation.
Resolving the driving question and using Kenyan contexts	Clearly distinguishes size/shape from orientation, answers the driving question in a connected argument, and gives accurate Kenyan examples and a thoughtful model revision.	Answers the driving question with some links between ideas; examples and model revision are accurate but brief.	Gives separate, unconnected statements; examples are missing or inaccurate and there is little model revision.

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